

Properties of a Linear Unbiased Minimum Variance Estimator

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Overview

A standard problem is to find the linear unbiased “minimum” variance estimator for the overdetermined problem[†]: $A\mathbf{x} = \mathbf{b}$, where $A \in \mathbf{R}^{n \times m}$, $\mathbf{b} \in \mathbf{R}^n$ is a random observation with $E[\mathbf{b}] = A\mathbf{x}$ and covariance V (assumed invertible), and $\mathbf{x} \in \mathbf{R}^m$ is the deterministic parameter we wish to estimate. We look for a linear unbiased estimator for $\mathbf{x}$ of the form $\mathbf{x}^* = L\mathbf{b}$, where $L \in \mathbf{R}^{m \times n}$. Since $\mathbf{x}^*$ is unbiased we must have $E[\mathbf{x} - \mathbf{x}^*] = \mathbf{x} - LA\mathbf{x} = (I - LA)\mathbf{x}$. For this to vanish for every $\mathbf{x}$, we require $LA = I$. This implies that A and L have rank m . The covariance of $L\mathbf{b}$ is LVL^T . The problem now is to find the best L such that $LA = I$ and the covariance matrix, LVL^T , is minimized in some sense.

The problem may be restated as choosing L to “minimize” the norm of LVL^T subject to the constraint $LA = I$. The question is what norm does one use. The “natural” norm of a matrix is $\sqrt{\text{tr}(M^T M)}$. This follows from considering the standard basis in $\mathbf{R}^{n \times m}$ and using the usual norm of the vector space $\mathbf{R}^{n \times m}$. The matrix of interest, LVL^T , is positive definite symmetric; in this case, each of the “trace” expressions, $\sqrt[k]{\text{tr}(M^k)}$ ($k \in [1, \infty)$), is a norm for a positive definite matrix $M_{m \times m}$ (these are the *Schatten k -norms*.)

Do the Norms Matter?

The question is, do different choices of norms lead to a different solution, L ? To answer this, we solve the constrained minimization problem that results from each of these trace norms[‡]:

$$F(L; k) = \text{tr} \left((LVL^T)^k - (LA - I)\Lambda_k^T \right) \quad k \in [1, \infty)$$

where $\Lambda_k \in \mathbf{R}^{m \times m}$ is a Lagrange multiplier.

The derivative of F with respect to L is given by:

$$DF(L)(H) = \text{tr} \left[k(LVL^T)^{k-1}(HVL^T + LVH^T) - H\Lambda_k^T \right]$$

At a minimum value, L^* , the derivative must vanish; therefore,

$$\text{tr} \left[k(L^*VL^{*T})^{k-1}(HVL^{*T} + L^*VH^T) - H\Lambda_k^T \right] = 0 \quad \forall H \in \mathbf{R}^{m \times n} \quad (1)$$

Before going further we list some facts regarding the trace inner product that will be used in the rest of the paper.

[†] This means that $n > m$.

[‡] Equivalently, we minimize the k^{th} power of the k^{th} trace norm.

- $\forall H \in \mathbf{R}^{n \times n}, H \text{ anti-symmetric} \Rightarrow \text{tr}(H) = 0.$
- Given a matrix $A \in \mathbf{R}^{m \times n}: \forall H \in \mathbf{R}^{m \times n}, \text{tr}(A^T H) = 0 \Rightarrow A \equiv 0.$
- $\forall A, H \in \mathbf{R}^{n \times n}, A \text{ symmetric} \wedge H \text{ anti-symmetric} \Rightarrow \text{tr}(A^T H) = 0.$
- Given a matrix $A \in \mathbf{R}^{n \times n}: \forall H \in \mathbf{R}^{n \times n}, H \text{ anti-symmetric} \wedge \text{tr}(A^T H) = 0 \Rightarrow A \text{ is symmetric}.$
- Given a matrix $A \in \mathbf{R}^{n \times n}: \forall H \in \mathbf{R}^{n \times n}, A \text{ symmetric} \wedge H \text{ symmetric} \wedge \text{tr}(A^T H) = 0 \Rightarrow A \equiv 0.$

The first fact follows from a property of square matrices; namely, $\text{tr}(A) = \text{tr}(A^T)$; and the definition of an anti-symmetric A , $A^T = -A$. The second fact follows since any inner product on a vector space, V , satisfies: $\forall y \in V, \langle x, y \rangle = 0 \Rightarrow x \equiv 0$. The third fact follows – using Fact 1 (or rather its underlying identity $\text{tr}(M) = \text{tr}(M^T)$) – as follows. If S and W are symmetric and anti-symmetric matrices in $\mathbf{R}^{n \times n}$, then $\text{tr}(SW) = \text{tr}((SW)^T) = \text{tr}(W^T S^T) = -\text{tr}(WS) = -\text{tr}(SW)$. This implies that $\text{tr}(SW) = 0$. The fourth fact follows from the unique decomposition of any matrix $A \in \mathbf{R}^{n \times n}$ as a sum of a symmetric and an anti-symmetric matrix: $A = A^s + A^w$, so $A^T = A^s - A^w$. For any anti-symmetric H , $\text{tr}(A^T H) = \text{tr}(A^s H) - \text{tr}(A^w H) = -\text{tr}(A^w H)$ by Fact 3. So if $\text{tr}(A^T H) = 0$ for all anti-symmetric H , then $\text{tr}(A^w H) = 0$ for all such H . Setting $H = A^w$ gives $\text{tr}((A^w)^2) = -\text{tr}((A^w)^T A^w) = -\|A^w\|_F^2 = 0$, so $A^w \equiv 0$ and A is symmetric. The fifth fact follows by setting H to A . Again, the fact that the trace inner product induces a norm implies that $A = 0$.

Continuing with the derivation, in the special case $k = 1$, equation (1) implies that $2VL^{*T} = A\Lambda_1^T$; or,

$$2L^{*T} = V^{-1}A\Lambda_1^T \quad (2)$$

Multiplying both sides by A^T and using the fact that $A^T L^{*T} = I$ yields $2I = A^T V^{-1} A \Lambda_1^T$. Since A has rank m , $A^T V^{-1} A$ is invertible, so $\Lambda_1^T = 2(A^T V^{-1} A)^{-1}$ (the right-hand side is symmetric, so $\Lambda_1 = \Lambda_1^T$). Combining this with equation (2) gives

$$L^* = (A^T V^{-1} A)^{-1} A^T V^{-1}$$

We now proceed with the case when $k > 1$. Given any $\tilde{H} \in \mathbf{R}^{m \times m}$, let $H = \tilde{H} A^T V^{-1}$. Then equation (1) becomes:

$$\text{tr} \left[k(L^* V L^{*T})^{k-1} (\tilde{H} A^T L^{*T} + L^* A \tilde{H}^T) - \tilde{H} (A^T V^{-1} A) \Lambda_k^T \right] = 0 \quad \forall \tilde{H} \in \mathbf{R}^{m \times m}$$

Or,

$$\text{tr} \left[k(L^* V L^{*T})^{k-1} (\tilde{H} + \tilde{H}^T) - \tilde{H} (A^T V^{-1} A) \Lambda_k^T \right] = 0 \quad \forall \tilde{H} \in \mathbf{R}^{m \times m} \quad (3)$$

since $L^* A = I$ and $A^T L^{*T} = I$.

Equation (3) holds for all $\tilde{H} \in \mathbf{R}^{m \times m}$. Let $M = (L^* V L^{*T})^{k-1}$ and $C = A^T V^{-1} A$. Since M is symmetric, $\text{tr}(M(\tilde{H} + \tilde{H}^T)) = 2 \text{tr}(\tilde{H} M)$ by cyclic invariance, so (3) reads

$$\text{tr} [\tilde{H} (2kM - C \Lambda_k^T)] = 0 \quad \forall \tilde{H} \in \mathbf{R}^{m \times m}.$$

By Fact 2, $2kM - C \Lambda_k^T = 0$. Since A has rank m , $(A^T V^{-1} A)^{-1}$ exists, and we can solve for Λ_k^T :

$$\Lambda_k^T = 2k (A^T V^{-1} A)^{-1} (L^* V L^{*T})^{k-1}.$$

Equation (1) now becomes:

$$\text{tr} \left[k(L^*VL^{*T})^{k-1}(HVL^{*T} + L^*VH^T) - 2kHA(A^TV^{-1}A)^{-1}(L^*VL^{*T})^{k-1} \right] = 0 \quad \forall H \in \mathbf{R}^{m \times n}$$

Or,

$$\text{tr} \left[\left((HVL^{*T} + L^*VH^T)/2 - HA(A^TV^{-1}A)^{-1} \right) (L^*VL^{*T})^{k-1} \right] = 0 \quad \forall H \in \mathbf{R}^{m \times n}$$

Which may be written

$$\text{tr} \left[\left(HVL^{*T} + (L^*VH^T - HVL^{*T})/2 - HA(A^TV^{-1}A)^{-1} \right) (L^*VL^{*T})^{k-1} \right] = 0 \quad \forall H \in \mathbf{R}^{m \times n}$$

Since $(L^*VH^T - HVL^{*T})/2$ is anti-symmetric and $(L^*VL^{*T})^{k-1}$ is symmetric, the trace of their product is zero. Therefore we may write the last equation as

$$\text{tr} \left[\left(HVL^{*T} - HA(A^TV^{-1}A)^{-1} \right) (L^*VL^{*T})^{k-1} \right] = 0 \quad \forall H \in \mathbf{R}^{m \times n}$$

Or,

$$\text{tr} \left[H \left(VL^{*T} - A(A^TV^{-1}A)^{-1} \right) (L^*VL^{*T})^{k-1} \right] = 0 \quad \forall H \in \mathbf{R}^{m \times n}$$

This implies that $(VL^{*T} - A(A^TV^{-1}A)^{-1})(L^*VL^{*T})^{k-1} = 0$. Since L^* has rank m , $(L^*VL^{*T})^{-(k-1)}$ exists, so that $VL^{*T} - A(A^TV^{-1}A)^{-1} = 0$. Hence,

$$L^* = (A^TV^{-1}A)^{-1}A^TV^{-1}$$

Therefore, it does not matter which of the matrix norms we use: the “best” linear unbiased estimator is the same.

Best Estimator under the Spectral Norm

We now use the results of the trace norms above to show that the estimator obtained has the smallest maximum eigenvalue over all linear estimators.

Lemma 1. *If Ω is a closed set in $\mathbf{R}^m$ and if $x^* \in \Omega$ is a common solution to $\min_{\{x \in \Omega: g(x)=0\}} \varphi_k(x)$ for every $k \in [1, \infty)$; further, if φ_k are functions which converge point-wise to the continuous function φ on Ω with the coercive property $\varphi(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$, then $x^* \in \text{argmin}_{\{x \in \Omega: g(x)=0\}} \varphi(x)$.*

Proof.

Existence: The set $\{x \in \Omega : g(x) = 0\}$ is non-empty. Let $\{x_n\}$ be a minimizing sequence in this set; that is, $\varphi(x_n)$ converges to the infimum. By the coercive property of φ , $\{x_n\}$ is bounded. Since Ω is closed, a subsequence converges to some $x' \in \Omega$. By the continuity of φ on Ω , $\varphi(x')$ equals the infimum, which is therefore finite and attained.

We proceed to show that x^* is a solution. Let $\tilde{x}$ be a solution of the problem $\min_{\{x \in \Omega: g(x)=0\}} \varphi(x)$; then for any $\epsilon > 0$, we have, for sufficiently large n , (since φ_n converges point-wise to φ)

$$\varphi_n(\tilde{x}) < \varphi(\tilde{x}) + \epsilon$$

Since x^* is a solution for φ_n , we also have

$$\varphi_n(x^*) \leq \varphi_n(\tilde{x})$$

Combining the two gives

$$\varphi_n(x^*) \leq \varphi_n(\tilde{x}) < \varphi(\tilde{x}) + \epsilon$$

Or,

$$\varphi_n(x^*) \leq \varphi(\tilde{x}) + \epsilon$$

Taking the limit as $n \rightarrow \infty$ we have:

$$\varphi(x^*) \leq \varphi(\tilde{x}) + \epsilon$$

Since this is true for all $\epsilon > 0$ we necessarily have that

$$\varphi(x^*) \leq \varphi(\tilde{x})$$

Noting that x^* satisfies the constraint, $g(x) = 0$, we infer that x^* is also a solution.

Theorem 1. *The linear unbiased estimator previously obtained for the least squares problem, $L^*\mathbf{b} = (A^T V^{-1} A)^{-1} A^T V^{-1} \mathbf{b}$, is also the solution that minimizes the maximum eigenvalue of the estimator's covariance matrix, $L V L^T$, subject to the constraint $L_{m \times n} A_{n \times m} = I_{m \times m}$.*

Proof. We apply Lemma 1 with the variable $L \in \mathbf{R}^{m \times n}$, $\Omega = \{L \in \mathbf{R}^{m \times n} : LA = I\}$ (closed), $\varphi_n(L) = \sqrt[n]{\text{tr}((L V L^T)^n)}$ (the n^{th} Schatten norm of $L V L^T$), and $\varphi(L) = \lambda_{\max}(L V L^T)$ (the spectral norm of $L V L^T$, which is positive definite since L has full row rank and V is positive definite). The constraint $g(L) = 0$ is vacuous since Ω already encodes $LA = I$. Theorem 1 follows after collecting the facts:

- For a positive definite matrix with eigenvalues $\lambda_i > 0$, $\sqrt[k]{\sum_i \lambda_i^k} \rightarrow \max_i \lambda_i$ as $k \rightarrow \infty$.
- The L_2 matrix norm of a positive definite matrix is its maximum eigenvalue.
- The L_2 norm is coercive.